



Hybrid solar-thermoelectric system by MIT

and Nenad Miljkovic, graduate student in mechanical engineering, combined the techniques of solar water heating and thermoelectric device for electricity generation to design a hybrid system. Their system is based on a standard solar energy collector called a parabolic trough. In this type of device, a long parabolic mirror reflects sunlight onto a fluid-filled tube that runs at its focal point. A mechanical pump moves the heated working fluid through the tube to the point of consumption. There, it can be used directly for space heating or for running industrial processes, or to produce steam for driving an electricity-generating turbine.

\$20 Raspberry Pi mini PC to come soon

When the makers of Raspberry Pi thought of making a low-cost computing device, they began by fixing a price point. They decided that they would make a sub-\$25 programming device, which will be affordable for students across the world. And they ended up with Raspberry Pi.

Eben Upton, executive director, Raspberry Pi Foundation, believes that the cost of the device is its biggest advantage. But he goes on to accept that he could have added more features had the cost been not a restraint.

Having said that, Upton is looking forward to bring a new type of Raspber-

ry Pi board at a much more economical price of \$20. "We're doing the Model A for \$25 next year - I imagine in two-three years' time it will be possible to do a board for \$20. But it will be difficult to get below \$20, even if you make various components for free," Upton was quoted as saying in an IT PRO report. The \$20 Raspberry Pi credit-card-sized mini CPU is expected to be available by 2015.

World's simplest clock made using one atom

How often does it happen that your PC, tablet, mobile phone and clocks in your house show the same time? Well, normally it is a rarity. While we push the question of how accurate time can be, here comes an end to the debate how simple can time be! Researchers at the University of California, Berkeley, have developed a clock with a single cesium atom.

The primary issue in making a clock from a particle of matter is that its oscillation frequency should be so high that one should never be able to measure it. To overcome the issue, researchers depended on the time dilation phenomenon, which basically means that as objects move far and come closer to a location, they experience less elapsed time in comparison to objects that are on one location the entire time.

So what did the researchers exactly do? "We essentially split an atom into

two halves, and had one stay where it is and the other go forward and come back," said physicist Hogler Muller of the University. "A tiny, tiny bit less time elapsed for the half that moved, so it oscillated less."

The fact that one half of the atom oscillated less than the other meant that when these halves are reunited, they did not recombine perfectly, but interference occurred that the scientists could measure. By knowing the size of this discrepancy and the extent to which the researchers disturbed the atom, the researchers could deduce the original frequency at which the atom oscillated, reported *Livescience*.

The moving half of the atom took about a third of a second less than the other half to make its round trip. Each half of the atom made about 10^{25} oscillations—a '1' with 25 zeroes behind it, equal to 10 trillion trillion—but the moving half made about 100,000 fewer oscillations than the still half, the report further stated.

Paper-thin tablet

Researchers at Intel, Plastic Logic and a Canadian college have come up with a tablet called PaperTab that is as thin as a piece of paper. It can be used either like a tablet or an e-book, with users just bending the paper-thin, flexible plastic screen to change the display. The device was recently showed off at the Consumer Electronics Show (CES), powered by a second-gen Intel Core i5 processor and featuring a fully interactive 27.2cm (10.7-inch) plastic touchscreen display.

This tablet is expected to ultimately replace the need for paper. "Plastic Logic's flexible plastic displays are completely transformational in terms of product interaction," Ryan Brotman, research scientist at Intel, said. "They allow a natural human interaction with electronic paper—being lighter, thinner and more robust than today's standard glass-based displays."

"Using several PaperTabs makes it much easier to work with multiple documents," said Roel Vertegaal, director of Queen's University's Human Media Lab.